**Department of Statistics & Computer Science**
**CSC3173: Artificial Intelligence (2024/25)**
**Programming Assignment 02**

**Due: On or before 15/11/2026 11.59pm**

In this assignment, you are required to design a **Financial Document Analysis Agent** based on the Medical Document (Report Analysis) Agent architecture demonstrated in class.

You may reuse the general structure of the Medical Document Agent, but the domain-specific components must be replaced or modified.

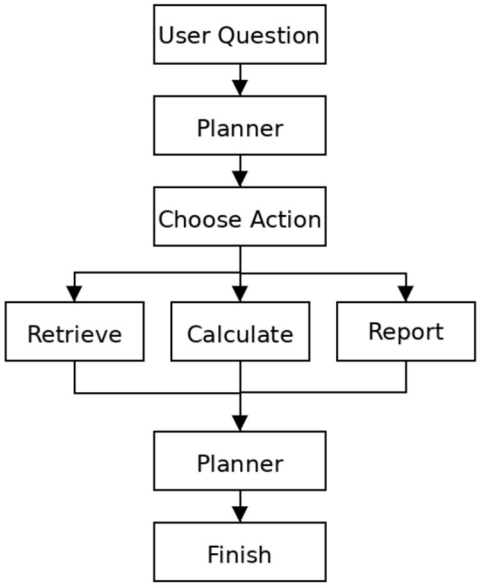
**Dataset:**
You are provided with three Apple annual report (10-K) PDF documents for 2023, 2024, and 2025. These documents must be used as the knowledge source for your Financial Document Agent.

**Financial Document Analysis Agent**
The analyst should be able to ask questions such as:

"How did the company's total revenue change between 2023 and 2025?"

The agent should automatically perform the necessary steps to answer the question.

A possible workflow is:




**Agentic Components**
First, identify the main agentic components in the Medical Document Agent demonstrated in class. Your identification should include the agent goal, state, planner, tools, actions/nodes, routing or iteration, and termination condition. Then, consider how these components can be modified or adapted when developing your own Financial Document Analysis Agent.

Your application should contain the following major stages.

A. **Document Processing**
B. **Semantic Retrieval**
C. **Financial Retrieval Tool**
D. **Financial Calculation Tool**

The LLM should not be relied upon to perform exact financial arithmetic. Create at least one Python-based calculation tool.

Your tool should support at least:

- Percentage change between two selected years

  Percentage change (%) = ((V_new - V_old) / V_old) x 100

You may additionally calculate quantities such as:

- absolute revenue change;
- product/category contribution;
- regional contribution;
- highest-growing category;
- highest/lowest-performing segment.

E. **Report Generation Tool**

****Agent Workflow Requirements :**

**The final system must include an agent state, a planner that selects the next action, tool/action nodes, conditional routing, iterative execution, and a termination condition. The planner should be able to direct the agent through actions such as RETRIEVE, CALCULATE, REPORT, and FINISH.**

**Deliverables:**

**1. Output Report:** The output report generated by your Financial Document Analysis Agent for the following test questions:

**Q1.** How did the company's total revenue change between 2023 and 2025?

**Q2.** Which product category contributed the most revenue in the latest year?

**Q3.** Create your own financial-analysis question. The question should require the agent to use information from the documents rather than answering only from general knowledge.

**2. Code:** Jupyter/Colab notebook (.ipynb) containing the complete Financial Document Analysis Agent

**Note:** Use the supplied Apple 10-K documents as the primary knowledge source for the required test questions.